

Medical Insight Explorer Agent

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Applied Data Scientist | Healthcare AI • Analytics • ML Systems

Bilingual conversational healthcare analytics with deterministic insights.

LangGraph + Gradio + Hugging Face

The screenshot shows a web browser window displaying the Hugging Face Spaces page for the 'Medical Insight Explorer Agent'. The browser's address bar shows the URL 'huggingface.co/spaces/Artur-Melnyk/Medical-Insight-Explorer-Agent'. The page header includes the 'Spaces' logo, the agent's name 'Artur-Melnyk/Medical-Insight-Exp...', a 'like' button with a count of 0, and status indicators for 'Agents' (Running) and 'App'. Navigation links for 'Files', 'Community', and 'Settings' are also present. The main content area features the title 'Medical Insight Explorer Agent' and a description: 'Conversational analytics over cleaned Medicare healthcare claims data. Konversationelle Analyse bereinigter Medicare-Healthcare-Claims-Daten.' It also states 'Deterministic analytics first. Persona-guided questions. Grounded explanations and optional visual outputs.' Below this, there are two dropdown menus: 'Language / Sprache' set to 'English' and 'Stakeholder persona / Stakeholder-Persona' set to 'Hospital Operations Analyst'. A description of the agent's capabilities follows: 'Explore provider utilization, claim volume, and reimbursement patterns.' Under the heading 'Recommended questions for this persona', there are three buttons: 'Show top inpatient providers', 'Show inpatient claims by state', and 'Show reimbursement distribution'. At the bottom, there is a text input field with the placeholder 'Ask a healthcare analytics question / Frage zur Healthcare-Analyse'.

huggingface.co/spaces/Artur-Melnyk/Medical-Insight-Explorer-Agent

Spaces | Artur-Melnyk/Medical-Insight-Exp... | like 0 | Agents | Running | App | Files | Community | Settings

Medical Insight Explorer Agent

Conversational analytics over cleaned Medicare healthcare claims data.
Konversationelle Analyse bereinigter Medicare-Healthcare-Claims-Daten.

Deterministic analytics first. Persona-guided questions. Grounded explanations and optional visual outputs.

Language / Sprache

English

Stakeholder persona / Stakeholder-Persona

Hospital Operations Analyst

Explore provider utilization, claim volume, and reimbursement patterns.

Recommended questions for this persona

Show top inpatient providers | Show inpatient claims by state | Show reimbursement distribution

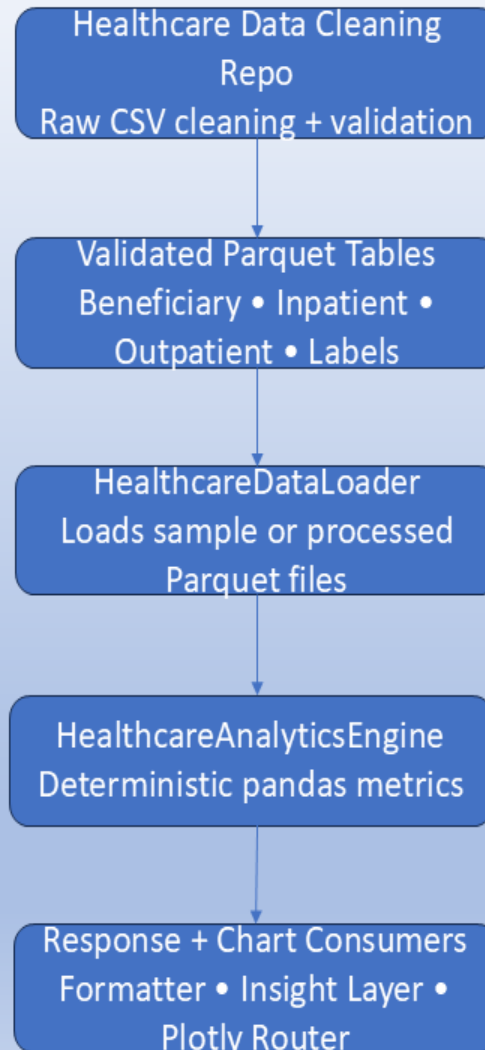
Ask a healthcare analytics question / Frage zur Healthcare-Analyse

Business Problem

- Healthcare claims data is large and difficult to explore
 - Analysts need grounded and deterministic outputs
- LLM systems in healthcare require explainability and safety
 - Goal: fast analytics without clinical advice

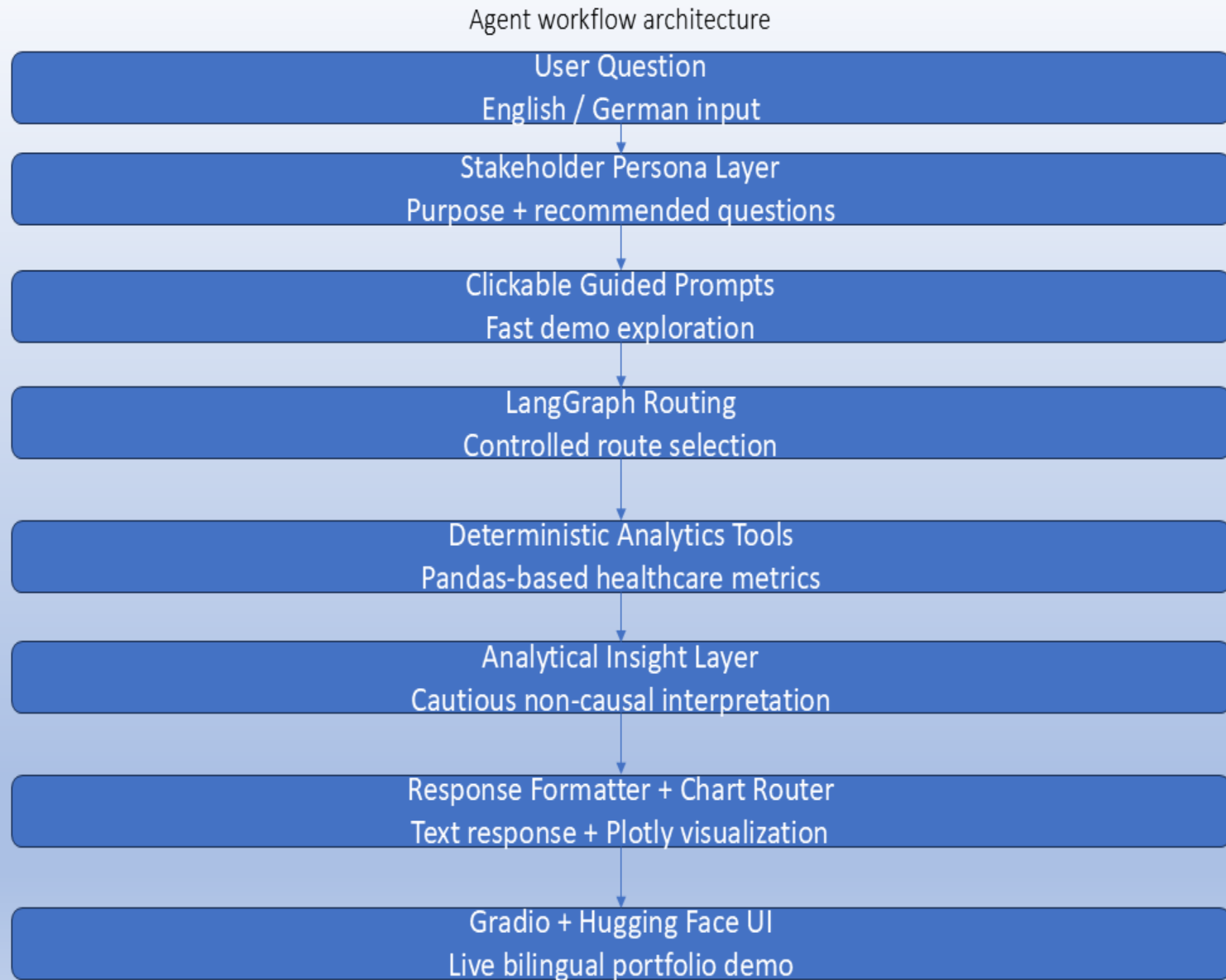
Project Ecosystem

Data Pipeline Architecture



Validated healthcare claims data flows into deterministic analytics before explanation or visualization.

Architecture



Persona-guided questions are routed into deterministic analytics, then explained through a controlled insight layer.

Technical Design Choices

Engineering decisions used to improve reliability, safety, and deployment stability

Design Decision	Why it was implemented
Compute First → Explain Second	Deterministic pandas analytics run before AI explanations to prevent fabricated healthcare statistics
Controlled Analytics Routing	User questions are routed only to predefined functions; the LLM never receives direct DataFrame access
Sample Data Deployment Strategy	Lightweight sample datasets were used to satisfy Hugging Face memory and runtime constraints
Localized Bilingual UX	English and German interaction improves accessibility and recruiter usability
Conditional Visualization Routing	Plotly charts are generated only for supported analytical routes to avoid misleading outputs
Persona-Guided Workflow	Stakeholder personas clarify who uses the app and why

Design principle: Safe computation → persona guidance → analytical insight → visualization

Product Evolution: From Generic Demo to Guided Analytics

Before:

- Generic healthcare analytics chatbot
- Example prompts without stakeholder context
- Outputs focused mostly on computed tables and charts
- Purpose was less clear for non-technical healthcare reviewers

After:

- Stakeholder personas clarify who uses the app
- Clickable guided prompts support faster demo exploration
- Analytical insight layer explains what results may indicate
- Safer claims-analytics framing avoids clinical overclaiming

Result: The app evolved from a technical demo into a persona-guided healthcare claims analytics assistant.

Key Features

- ✓ English/German UX
- ✓ Stakeholder personas
- ✓ Clickable guided prompts
 - ✓ Deterministic analytics
 - ✓ Analytical insight layer
 - ✓ Plotly charts
- ✓ LangGraph orchestration
- ✓ Hugging Face deployment

English Demo

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Language / Sprache

English

Stakeholder persona / Stakeholder-Persona

Healthcare Fraud Investigator

Review provider concentration, unusual claim volume, and reimbursement patterns.

Recommended questions for this persona

Show top outpatient providers

Show reimbursement distribution

What is the diabetes cost summary?

Ask a healthcare analytics question / Frage zur Healthcare-Analyse

Show reimbursement distribution

Run analysis / Analyse ausführen

Clear / Zurücksetzen

Agent response / Antwort des Agenten

Summary:

The inpatient reimbursement distribution was calculated from non-missing claim reimbursement values.

Analytical insight:

The reimbursement distribution can reveal skew, typical claim amounts, and high-cost outliers.

Computed result:

Reimbursement distribution was computed from non-missing `InscClaimAmtReimbursed` values. See the histogram below for the visual distribution.

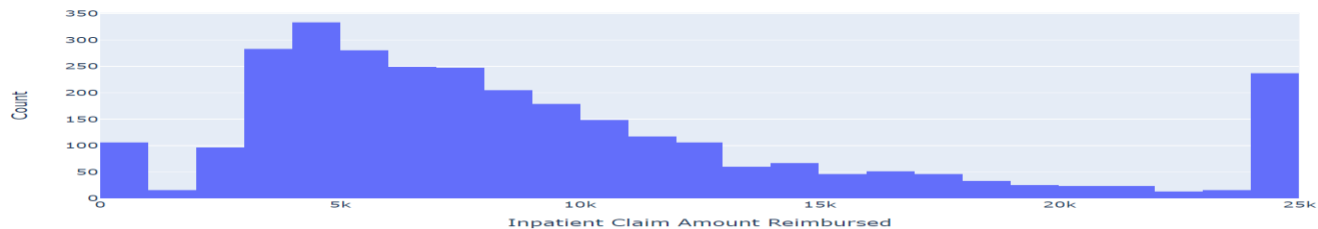
Method note:

This answer is based on deterministic pandas computations over cleaned healthcare claims data.

This is claims analytics only and does not provide diagnosis, treatment advice, or clinical recommendations.

Visualization output / Visualisierung

Distribution of Inpatient Claim Reimbursement



Supported languages: English | Deutsch

Deterministic analytics • Claims data only • No diagnosis

German Demo

Spaces

Artur-Melnyk/Medical-Insight-Exp...

like 0

Agents

Running

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Medical Insight Explorer Agent

Conversational analytics over cleaned Medicare healthcare claims data.
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Language / Sprache

Deutsch

Stakeholder persona / Stakeholder-Persona

Healthcare-Policy-Researcher

Analysiert populationsbezogene Nutzung, Beneficiary-Merkmale und regionale Trends.

Empfohlene Fragen für diese Persona

Wie hoch ist das durchschnittliche Alter der Beneficiaries?

Zeige stationäre Claims nach Bundesstaat

Zeige ambulante Claims nach Bundesstaat

Ask a healthcare analytics question / Frage zur Healthcare-Analyse

Wie hoch ist das durchschnittliche Alter der Beneficiaries?

Run analysis / Analyse ausführen

Clear / Zurücksetzen

Agent response / Antwort des Agenten

Zusammenfassung:
Das durchschnittliche Beneficiary-Alter beträgt ungefähr 83.46 Jahre.

Analytische Einordnung:
Die Altersübersicht unterstützt die Einordnung der Beneficiary-Population im Datensatz.

Berechnetes Ergebnis:
count: 7679
mean_age: 83.4564396405782
median_age: 84.0
min_age: 36.0
max_age: 111.0

Methodischer Hinweis:
Diese Antwort basiert auf deterministischen pandas-Berechnungen über bereinigte Healthcare-Claims-Daten.

Dies ist Claims-Analytik und keine Diagnose, Behandlungsberatung oder klinische Empfehlung.

Visualization output / Visualisierung

Keine Visualisierung für diese Frage erstellt



Diese Frage eignet sich besser für eine textbasierte Zusammenfassung.

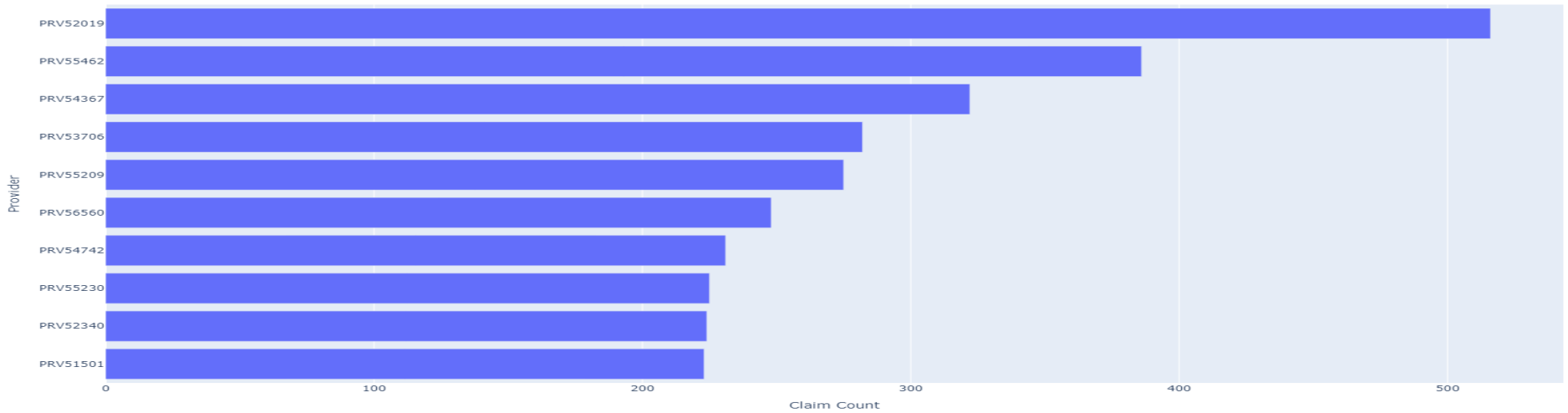
Supported languages: English | Deutsch

Deterministic analytics • Claims data only • No diagnosis

Use via API  • Built with Gradio 

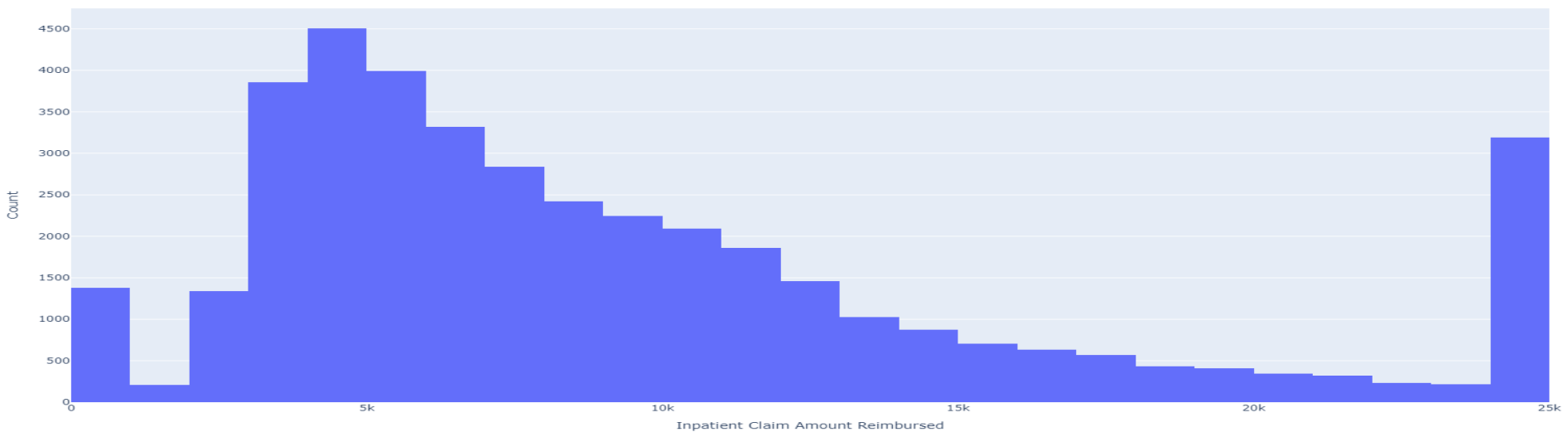
Analytics Outputs

Top 10 Inpatient Providers by Claim Count



Insight: Provider activity is concentrated among a small subset of institutions, which can support utilization monitoring and operational review.

Distribution of Inpatient Claim Reimbursement



Insight: Inpatient reimbursements are strongly right-skewed. Most claims are concentrated in lower-to-mid reimbursement ranges, while a smaller group of high-cost claims extends toward the upper cap.

Safety & Responsible AI

- ✓ Claims analytics only
 - ✓ No diagnosis
- ✓ No treatment recommendations
- ✓ Explainable deterministic calculations
 - ✓ Safer healthcare AI interaction
 - ✓ Personas guide workflows only
 - ✓ No clinical decision-making
 - ✓ No causal medical conclusions

Live Demo & Portfolio

Key Takeaways

- ✓ Persona-guided healthcare analytics
- ✓ Explainable deterministic workflow
 - ✓ Live Hugging Face deployment
 - ✓ Bilingual recruiter-ready UX

Hugging Face:

<https://huggingface.co/spaces/Artur-Melnyk/Medical-Insight-Explorer-Agent>

GitHub:

<https://github.com/ArturMelnyk-analyst/Medical-Insight-Explorer-Agent>

Built using: Python • Pandas • Plotly • Gradio • LangGraph • Hugging Face